Generative Drifting for One-Step Super-Resolution

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Abstract

This work investigates the applicability of the Generative Drifting framework to single-step image super-resolution. Existing super-resolution methods face a fundamental trade-off between perceptual quality and inference speed. To address this problem, we explore the Generative Drifting paradigm, which models generation as deterministic transport between probability distributions using a learned vector field. We propose a conditional formulation that learns a direct transport from low-resolution to high-resolution images in a single forward pass. Experiments show competitive reconstruction quality and improved efficiency compared to GAN-based and distilled diffusion models, measured using PSNR, SSIM, LPIPS, and inference time.

Keywords: Generative Drifting, Super-Resolution, One-Step Generation

1 Introduction

Generative modeling aims to learn mappings between complex high-dimensional data distributions. In image restoration tasks such as single-image super-resolution (SR), the objective is to reconstruct a high-resolution (HR) image from its low-resolution (LR) observation. Formally, this problem can be interpreted as learning a conditional transformation that maps images from the LR domain to the HR domain. Single-image super-resolution has numerous practical applications, including medical imaging, remote sensing, video restoration, and digital photography. Despite substantial progress in recent years, designing models that simultaneously achieve high perceptual quality and efficient inference remains a challenging problem.

Early deep learning approaches to super-resolution relied on convolutional neural networks trained with pixel-wise reconstruction losses. Work such as SRCNN [1] first demonstrated the effectiveness of deep learning for SR, while subsequent architectures such as VDSR [2], EDSR [3] and RCAN [4] further increased performance through deeper networks and advanced architectural designs. Such methods typically produce images with high peak signal-to-noise ratio (PSNR) but often fail to recover realistic textures.

To address this limitation, adversarial learning was introduced in SRGAN [5], which employs a discriminator network to encourage photo-realistic outputs. Subsequent improvements such as ESRGAN [6] further enhanced perceptual quality through improved architectures and perceptual loss functions. However, adversarial training is often unstable and may introduce artifacts or hallucinated structures that are inconsistent with the input image.

More recently, diffusion-based generative models have achieved remarkable results in image restoration. Methods such as SR3 [7] formulate super-resolution as an iterative denoising process and produce highly realistic reconstructions. Nevertheless, diffusion models typically require dozens or hundreds of inference steps, which significantly increases computational cost. To address this issue, several works attempt to accelerate diffusion sampling through distillation or architectural modifications [8]. For example, SinSR [9] and OSEDiff [10] approximate multi-step diffusion dynamics with a single-step network. Despite these advances, such methods often rely on complex training procedures or require pre-trained diffusion models.

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An alternative direction for fast generative modeling is based on transport and flow formulations. Normalizing flows have been applied to super-resolution in SRFlow [11], enabling probabilistic reconstruction of HR images via exact likelihood training. More recently, several works propose learning transport dynamics through vector fields, including Flow Matching [12] and related one-step generative modeling frameworks such as Consistency Models [13] and MeanFlows [14]. These approaches interpret generation as transporting samples between distributions along learned velocity fields.

A recent framework, Generative Drifting [15], proposes a deterministic generative model trained using a drifting vector field that moves generated samples toward the data distribution. The method defines attraction and repulsion forces between samples in feature space and learns a transport map that aligns generated and real data distributions. Unlike diffusion-based approaches, drifting-based models perform generation in a single forward pass, enabling efficient inference.

In this work, we investigate the applicability of the Generative Drifting framework to the conditional task of single-image super-resolution. While the original drifting formulation focuses on unconditional generative modeling, super-resolution requires conditioning on the observed low-resolution image. We therefore propose a conditional drifting formulation that learns a direct transport map from LR images to their HR counterparts.

The proposed method constructs a drifting vector field in a joint feature space that incorporates both residual image representations and conditioning information from the LR input. This formulation allows generated samples to move toward the manifold of realistic HR images while preserving consistency with the observed LR input. As a result, the model performs super-resolution in a single forward pass without iterative refinement or adversarial training.

The contributions of this work can be summarized as follows:

- We propose a conditional formulation of the Generative Drifting framework for the task of single-image super-resolution.
- We introduce a drifting vector field defined in a joint feature space that incorporates both residual image information and conditioning from the LR input.
- We develop a one-step super-resolution model that avoids iterative diffusion sampling and adversarial training while maintaining competitive reconstruction quality.
- We conduct experimental evaluation on standard super-resolution benchmarks using PSNR, SSIM, LPIPS, and runtime metrics.

The paper proposes a drifting-based method for conditional image super-resolution, providing efficient one-step reconstruction and distinguished by the use of a conditional drifting vector field defined in a joint feature space.

2 Problem Statement

Let

$$\mathcal{D} = \{(x_i^{LR}, x_i^{HR})\}_{i=1}^{N}$$

be a dataset of paired low-resolution and high-resolution images, where

$$x_i^{LR} \in \mathbb{R}^{h \times w \times c}, \qquad x_i^{HR} \in \mathbb{R}^{H \times W \times c},$$

and $H = sh$, $W = sw$ for a fixed scaling factor s.

The goal is to construct a model that reconstructs a high-resolution image from its low-resolution observation.

Let

$$f_\theta : \mathbb{R}^{h \times w \times c} \to \mathbb{R}^{H \times W \times c}$$

be a parametric family of functions, where $\theta \in \Theta$ is the vector of model parameters and Θ is the admissible parameter space. For an input image x_i^{LR}, the model predicts

$$\hat{x}_i^{HR} = f_\theta(x_i^{LR}).$$

The reconstruction is performed as a residual correction of an upsampled low-resolution image. Let

$$u : \mathbb{R}^{h\times w\times c} \to \mathbb{R}^{H\times W\times c}$$

be a fixed upsampling operator. Define

$$x_i^{UP} = u(x_i^{LR}),$$

and the generated and target residuals

$$r_i^{gen} = \hat{x}_i^{HR} - x_i^{UP}, \qquad r_i^{pos} = x_i^{HR} - x_i^{UP}.$$

Following the Generative Drifting framework, for each generated sample a drift vector

$$V_i = \sum_{j=1}^{N} W_{ij}^{pos} r_j^{pos} - \beta \sum_{j=1}^{N} W_{ij}^{neg} r_j^{gen}$$

is constructed, where W_{ij}^{pos} and W_{ij}^{neg} are attraction and repulsion weights, respectively, and $\beta > 0$ is a repulsion coefficient.

The drift vector defines a target point

$$\tilde{x}_i^{HR} = \hat{x}_i^{HR} + \eta V_i,$$

where $\eta > 0$ is the drift step.

To estimate the model parameters, consider the following loss terms.

The drift loss enforces consistency with the drift direction:

$$L_{drift}^{(i)} = \left\| \hat{x}_i^{HR} - \tilde{x}_i^{HR} \right\|_1 .$$

The reconstruction loss measures the deviation from the ground-truth high-resolution image:

$$L_{rec}^{(i)} = \left\| \hat{x}_i^{HR} - x_i^{HR} \right\|_1 .$$

The low-resolution consistency loss enforces compatibility between the reconstructed image and the observed low-resolution input:

$$L_{LR}^{(i)} = \left\| D(\hat{x}_i^{HR}) - x_i^{LR} \right\|_1 ,$$

where

$$D : \mathbb{R}^{H\times W\times c} \to \mathbb{R}^{h\times w\times c}$$

is a fixed degradation operator.

The total empirical loss is defined as

$$L(\theta) = \frac{1}{N} \sum_{i=1}^{N} \left(L_{drift}^{(i)} + \lambda_{LR} L_{LR}^{(i)} + \lambda_{rec} L_{rec}^{(i)} \right),$$

where $\lambda_{LR} > 0$ and $\lambda_{rec} > 0$ are weighting coefficients.

The optimal model parameters are obtained by solving the optimization problem

$$\theta^* = \arg\min_{\theta\in\Theta} L(\theta).$$

The solution θ^* defines the super-resolution model f_{θ^*}, and the inference procedure consists of a single forward pass

$$\hat{x}^{HR} = f_{\theta^*}(x^{LR}).$$

3 Preliminary experiment

Goal. The goal of the preliminary experiment is to verify the feasibility of conditional Generative Drifting for one-step super-resolution and compare it with a plain residual baseline.

Dataset. We use FashionMNIST (60k train / 10k test, grayscale 28×28). Low-resolution inputs are generated synthetically by Gaussian blur (kernel 5×5, $\sigma = 1.0$) followed by bicubic downsampling with scale factor 4 and bicubic upsampling.

Setup. The model is a lightweight residual CNN (6 blocks, 64 channels). The drifting version additionally uses a frozen feature encoder. Both models are trained for 3 epochs with AdamW, learning rate $2 \cdot 10^{-4}$, batch size 128. Evaluation metric is PSNR.

Results. Bicubic interpolation achieves 15.548 dB PSNR, the plain model reaches 18.619 dB, and the drifting model achieves 18.426 dB. Thus, both models outperform bicubic interpolation, while drifting does not improve over the plain baseline in this toy setup.

Preliminary report. The results confirm that the proposed formulation is feasible and can be trained stably. However, its advantage is not yet visible on this simple dataset, which motivates further experiments on more realistic benchmarks and with additional metrics.

Table 1: PSNR (dB) on FashionMNIST toy experiment

Method	PSNR
Bicubic	15.548
Plain SR	18.619
Drift SR	18.426

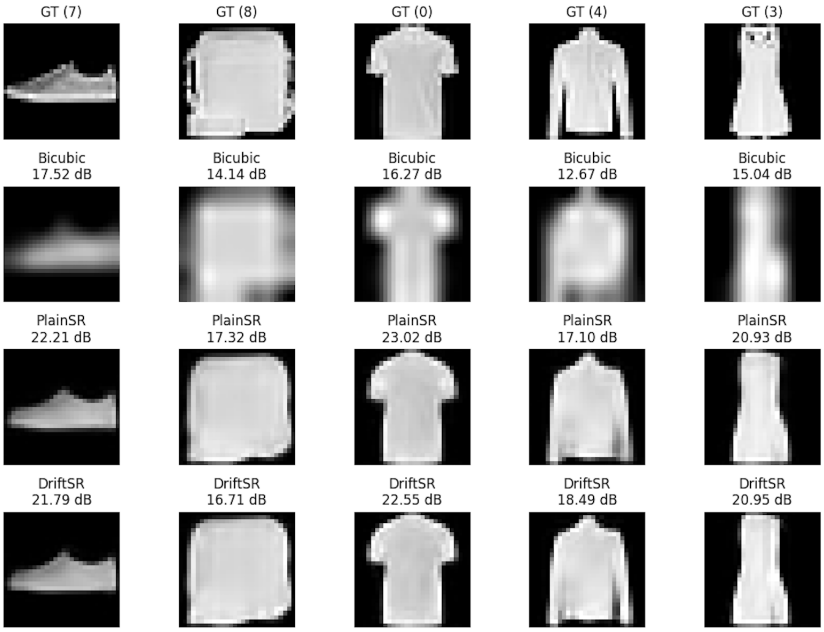

Figure 1: Qualitative comparison of super-resolution results. From top to bottom: ground truth, bicubic interpolation, plain SR, and drifting SR.

4 Discussion

TODO

5 Conclusion

TODO

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A Appendix / supplemental material

A.1 Additional experiments / Proofs of Theorems

TODO